MACHINE LEARNING WITH SCIKIT-LEARN

Classification & Ensembles

06

Open every remaining black box — logistic, trees, SVM — then let committees of models outvote any single one.

Master's Degree Program · Hands-on with Google Colab

AGENDA

Today's agenda

01 **Decision boundaries**
what a classifier actually draws

02 **Logistic regression**
sigmoid, probabilities, weights

03 **Decision trees**
rules you can read aloud

04 **Naive Bayes & SVM**
the 90-second tour

05 **Why ensembles**
one tree = opinion, many = wisdom

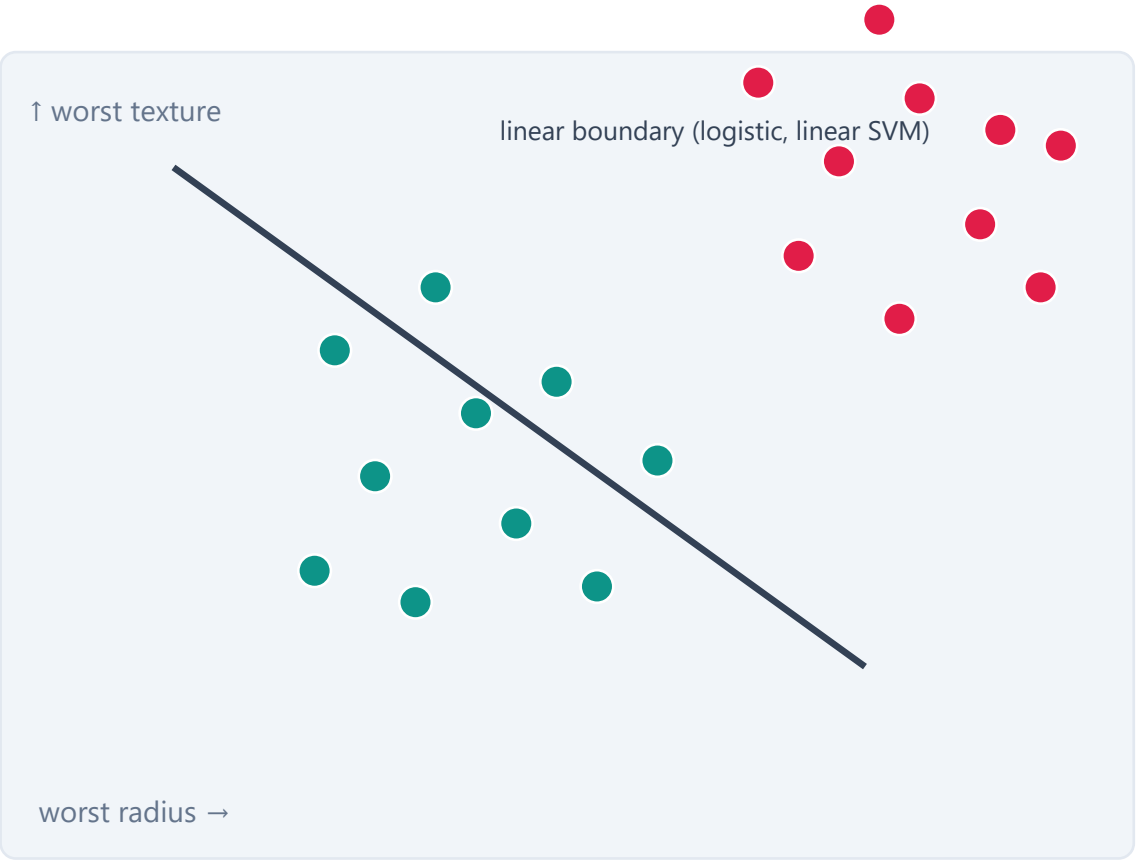
06 **Bagging → RandomForest**
+ feature importances

07 **Boosting → HistGradientBoosting**
and XGBoost beyond

08 **Voting & the full scoreboard**
who wins when

CONCEPT

A classifier draws a boundary

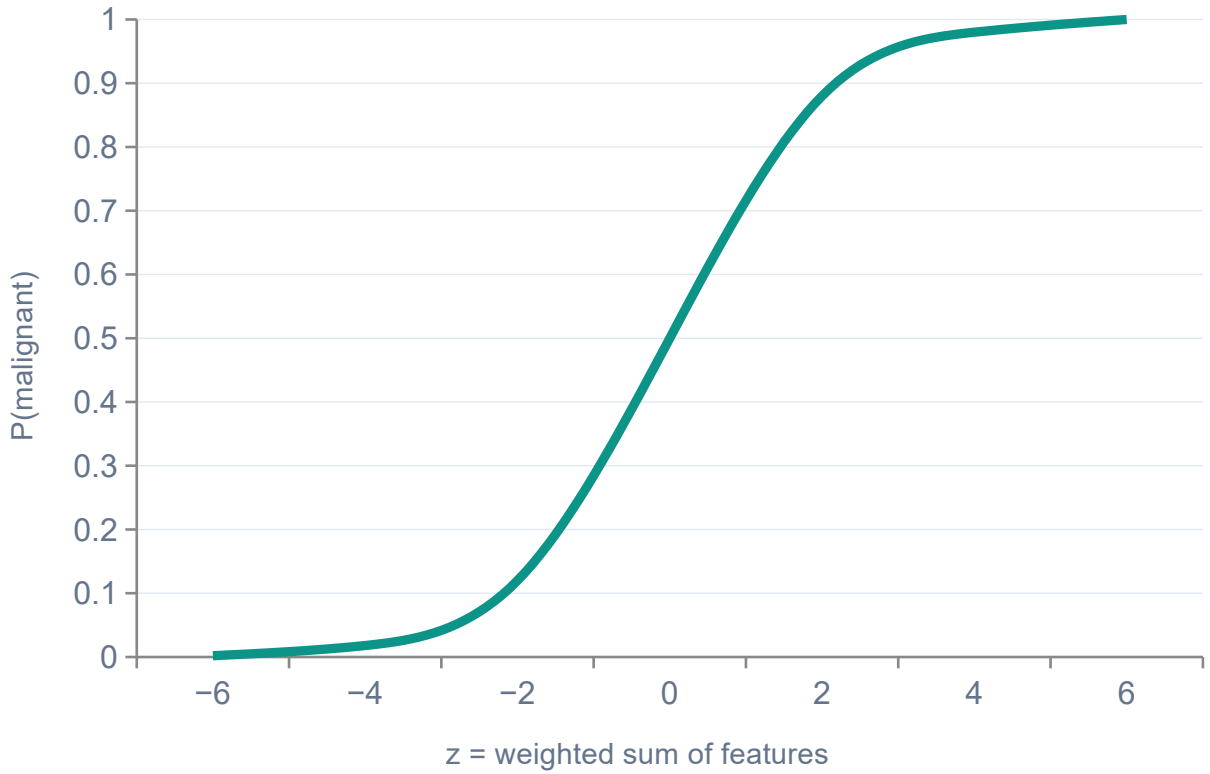


● benign ● malignant (breast cancer, 2 of 30 features)

- **Same goal, different geometry**
- logistic / linear SVM → straight line (hyperplane)
- KNN → wiggle around neighbors
- tree → axis-aligned rectangles
- kernel SVM → curved in original space
- **All of them output either a class (predict) or a probability (predict_proba)**

BLACK BOX #2

Logistic regression: a probability pump



- **z = w·x + b (same as linear!)**
- sigmoid squashes z into probability 0–1
- **predict_proba → the number**
- predict → threshold at 0.5 by default
- **"Logistic regression" = classification. The name lies.**

```
CV accuracy 0.974 · ROC-AUC 0.995
```

INTERPRETATION

Logistic weights = evidence, in log-odds

feature (standardized)	weight	reads as
worst texture	−1.25	coarser texture → malignant
radius error	−1.07	size variability → malignant
worst symmetry	−0.96	asymmetry → malignant
worst concave points	−0.94	concavities → malignant
area error	−0.94	area variability → malignant

(Signs here are data-specific; magnitude = strength of evidence. Positive class = benign in this encoding.)

- **Same reading skill as S05**
- linear: weights move the VALUE
- logistic: weights move the PROBABILITY
- exp(w) = odds ratio — "evidence multiplier"
- **Trains fast, scales big — the industry default first classifier**

BLACK BOX #3

Decision trees: 20 questions for tumors

"worst radius ≤ 16.8?"
no → malignant-ish branch

→

"concave points ≤ 0.14?"
yes → likely benign

→

"area error ≤ 91.6?"
refine again

→

leaf
predict class + proba

- **How is a question chosen?**
- try candidate thresholds per feature
- keep the split that best separates the classes (Gini / entropy — purity math, intuition is enough)
- Stop when: max_depth / min_samples_leaf / pure leaves
- No scaling needed — thresholds compare raw values

INTERPRETATION

A real fitted tree, read aloud

export_text(tree, feature_names=...) — real fitted tree

```
|--- worst radius <= 16.80
|   |--- worst concave points <= 0.14
|   |   |--- area error <= 91.56
|   |   |   |--- class: 1   (benign)
|   |   |--- area error >  91.56
|   |   |   |--- class: 0   (malignant)
|   |--- worst concave points >  0.14
|   |   |--- worst texture <= 25.62
|   |   |   |--- class: 1
|   |--- ...
|--- worst radius >  16.80
|   |--- texture error <= 0.47
|   |   |   |--- class: 1
|   |--- ... depth 3, test accuracy 0.944
```

- **This IS the model**
- export_text(tree, feature_names=...)
- a doctor can audit every decision
- "Small, smooth tumors are benign" — the data says so

  – plot_tree() draws it as a flowchart

Unleashed, a tree memorizes everything

tree	train acc	CV acc	AUC
unrestricted	1.000	0.910	0.900
max_depth = 3	0.977	0.924	0.934

- **A free tree grows until every leaf is pure**
- one patient per leaf → 100% train accuracy
- **its CV accuracy (0.910) is the WORST of today's models**
- restrict depth / min_samples → generalizes better
- **...or grow many restricted trees and average them → RandomForest (soon)**

Also: trees are unstable

Two random samples → two quite different trees.

High variance is the disease.
Ensembles are the cure.

Remember this slide — the next one pays it off.

Naive Bayes & SVM

GaussianNB — 0.930 acc / 0.986 AUC

Assumes features are independent given the class (naive — they never are).

And still works shockingly well.

· trains in a blink, tiny data OK
· THE classic for spam & text (S06 of any NLP course)

Use when: speed, small data, or a strong baseline.

SVM (RBF) — 0.977 acc / 0.995 AUC ⋆ today's best single

Draws the widest street between classes (max margin), possibly curved via kernels.

· powerful on small/medium clean data
· NEEDS scaling (S03!)
· doesn't scale to millions of rows; no predict_proba by default

Use when: medium data, strong boundaries.

SCOREBOARD 1/2

Single models, stratified 5-fold CV

model	CV acc	AUC
Dummy (most_frequent)	0.627	0.500
DecisionTree (free)	0.910	0.900
GaussianNB	0.930	0.986
KNN (k=5)	0.963	0.985
LogisticRegression	0.974	0.995
SVM (RBF)	0.977	0.995

- **Every model beats Dummy by miles — but AUC (0.900 vs 0.995) separates them far better than accuracy, and ranks the free tree LAST**

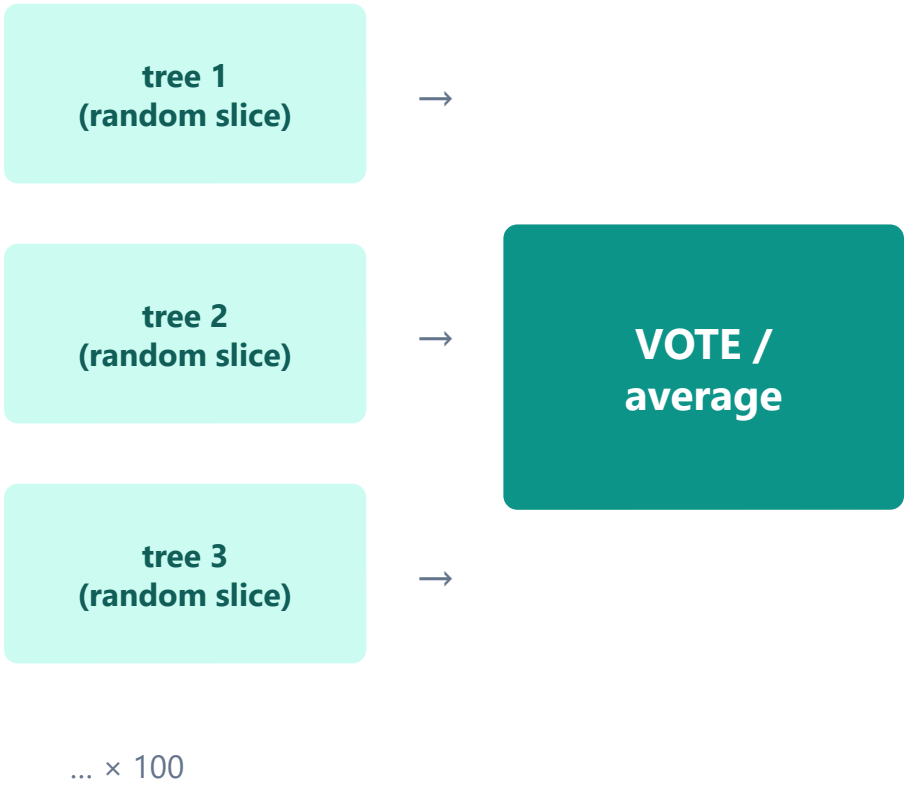
Next question

Can we COMBINE models —
take a few imperfect opinions
and beat the best single one?

(The answer is yes —
and it's how real tabular ML
competitions are won.)

Wisdom of crowds: why committees work

- **One tree: high variance**
- new sample → new tree → new opinions
- **Ask 100 slightly-different trees**
- each sees a random slice of rows & features
- errors are UNCORRELATED-ish → they cancel in the vote
- **Conditions (the ML version of Galton's ox):**
  - each member better than chance
  - members disagree with each other
  - aggregate by voting (classify) or averaging (regress)



RandomForest: bagging + a random twist

bootstrap rows
sample WITH replacement
→
random features per split
the "forest" twist
→
grow deep tree
no pruning — variance is fine
→
× 100 trees → vote
aggregate

model	CV acc	AUC
single free tree	0.910	0.900
RandomForest(100)	0.956	0.989

- **Feature importances (real, this dataset)**
- worst area 0.15 · concave points 0.13 · mean concave 0.11
- ≈ "which columns did the forest actually use?"
- Caveat: correlated features share/split credit — read as a hint, not a verdict

Why it works

Deep trees = low bias, high variance. Averaging 100 uncorrelated-ish trees ≈ variance ÷ 100.

Strong default: little tuning, no scaling needed, handles mixed types.

Boosting: each tree fixes the last one's mistakes

tree 1
predicts; notes its misses

→

tree 2
focuses on tree 1's misses

→

tree 3…N
keeps chasing residuals

→

sum of all
shallow but cumulative

model	CV acc	AUC
RandomForest(100)	0.956	0.989
HistGradientBoosting	0.967	0.992

- **Bagging: parallel, deep trees, vote · Boosting: sequential, shallow trees, add up — HistGB is sklearn's fast boosting (native NaN support!)**
- **Same family beyond sklearn: XGBoost, LightGBM, CatBoost — the kings of tabular Kaggle. Your S08 project can try them.**

Voting & stacking: mix different species

```
vote = VotingClassifier([
    ("lr",  make_pipeline(StandardScaler(),
                          LogisticRegression(max_iter=5000))),
    ("rf",  RandomForestClassifier(random_state=42)),
    ("knn", make_pipeline(StandardScaler(),
                          KNeighborsClassifier()))],
    voting="soft")   # average PROBABILITIES
```

- **hard voting**
- each model votes a class label
- **soft voting (better)**
- average predict_proba — confidence counts
  - Stacking: a small "meta-model" learns to weight the members — sklearn's StackingClassifier
- **Members should DISAGREE (diversity > quantity)**

model	CV acc	AUC
Voting(LR + RF + KNN)	0.972	0.995

SCOREBOARD 2/2

The full table — and its three lessons

model	CV acc	AUC
Dummy (most_frequent)	0.627	0.500
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LogisticRegression	0.974	0.995
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- **Lesson 1**
- ensembles lifted the tree 0.910 → 0.956+
- **Lesson 2**
- on this easy set, tuned singles match ensembles —
- no free lunch; ALWAYS run the table (S04 method)
- **Lesson 3**
- on bigger, messier tabular data, boosting usually wins
- (that's why S08 lets you try XGBoost)

GUIDE

Which classifier when?

situation	start with	strength	watch out
Baseline / speed / text	GaussianNB	instant	independence lie
Explanation: audit trail	Logistic / small tree	readable	linear-ish
Small-medium, clean	SVM (RBF)	top accuracy	scaling + scale-up
Tabular default	RandomForest	robust, no scaling	big memory
Max tabular accuracy	HistGB / XGBoost	Kaggle kings	tuning (S07)
squeeze out the last %	Voting / Stacking	diversity pays	complexity

And remember: the winner varies per dataset — the S04 comparison protocol is the real skill.

LAB 06 · DO IT IN GOOGLE COLAB

ประกวดโมเดลจัดหมวดมะเร็งเต้านม — รวมถึง **ensembles**

YOUR MISSION (โจทย์ภาษาไทยใน Colab)

- เทียบ Dummy / LogisticRegression / GaussianNB / SVM / KNN / DecisionTree ด้วย StratifiedKFold (acc + ROC-AUC)
- โชว์กฎของ DecisionTree(max_depth=3) ด้วย export_text แล้วอ่านออกเสียงเป็นประโยคไทย
- พิสูจน์ ensemble effect: tree เดี่ยว vs RandomForest vs HistGradientBoosting
- ดู feature_importances_ ของ RandomForest — ฟีเจอร์ไหนสำคัญที่สุด ตรงกับความรู้ทางการแพทย์ไหม
- โจทย์ชวนคิด: สร้าง VotingClassifier ของตัวเอง (เลือกสมาชิกให้หลากหลาย) แล้วดูว่าชนะสมาชิกตัวไหนไหม

Learning goal

Run today's full zoo on breast cancer with stratified CV, read real trees and importances, and prove the ensemble effect yourself.

Vibe coding corner

ลองถาม AI: “อธิบายความต่างระหว่าง bagging กับ boosting เหมือนสอนเด็ก 12 ขวบ” แล้วเขียนคำอธิบาย 3 บรรทัดเป็นภาษาไทยใน markdown cell — เช็กว่าตรงกับสไลด์ไหม

KEY TAKEAWAYS

1. Classifiers differ in the boundaries they draw: lines (logistic/SVM), rectangles (trees), neighborhoods (KNN).
2. Logistic turns evidence into probability; small trees turn it into rules you can audit.
3. A free tree memorizes (CV 0.910, worst of the day) — restriction or aggregation fixes it.
4. Bagging averages deep trees (RandomForest 0.956); boosting chains shallow ones (HistGB 0.967); voting mixes species (0.972).
5. No universal winner — the S04 protocol (same folds, right metric, Dummy anchor) is the real skill.

NEXT

Session 07 — Unsupervised Learning + Hyperparameter Tuning — structure, and squeezing